



A Project Report on
Color Blindness Detection using Eshihara Plates

Submitted by,

Prasad Ingole	(Exam Seat No. 202301040064)
Suyash Dahitule	(Exam Seat No. 202301040053)
Ketan Sonawane	(Exam Seat No. 202301040161)
Om Ranmode	(Exam Seat No. 202301040026)

Guided by,

Dr. Chetna Nemade

**A Report submitted to MIT Academy of Engineering, Alandi(D), Pune,
An Autonomous Institute Affiliated to Savitribai Phule Pune University
in partial fulfillment of the requirements of**

**THIRD YEAR BACHELOR OF TECHNOLOGY in
Computer Engineering**

Department of Computer Engineering

MIT Academy of Engineering

(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Alandi (D), Pune – 412105

(2025–2026)

CERTIFICATE

It is hereby certified that the work which is being presented in the Third Year Major Project-1 Report entitled “**Color Blindness Detection using Eshihara Plates**”, in partial fulfillment of the requirements for the award of the Bachelor of Technology in Computer Engineering and submitted to the **Department of Computer Engineering** of **MIT Academy of Engineering, Alandi(D), Pune, Affiliated to Savitribai Phule Pune University (SPPU), Pune**, is an authentic record of work carried out during Academic Year **2025–2026** Semester **V**, under the supervision of **Dr. Chetna Nemade, Department of Computer Engineering**

Prasad Ingole (Exam Seat No. 202301040064)

Suyash Dahitule (Exam Seat No. 202301040053)

Ketan Sonawane (Exam Seat No. 202301040161)

Om Ranmode (Exam Seat No. 202301040026)

Dr. Chetna Nemade
Project Advisor

Dr. Kanchan Dhote
Project Coordinator

Dr. Pramod Ganjewar
HoD

External Examiner

DECLARATION

We the undersigned solemnly declare that the project report is based on our own work carried out during the course of our study under the supervision of **Dr. Chetna Nemade**.

We assert the statements made and conclusions drawn are an outcome of our project work. We further certify that

1. The work contained in the report is original and has been done by us under the general supervision of our supervisor.
2. The work has not been submitted to any other Institution for any other degree/diploma/certificate in this Institute/University or any other Institute/University of India or abroad.
3. We have followed the guidelines provided by the Institute in writing the report.
4. Whenever we have used materials (data, theoretical analysis, and text) from other sources, we have given due credit to them in the text of the report and giving their details in the references.

Prasad Ingole (Exam Seat No. 202301040064)

Suyash Dahitule (Exam Seat No. 202301040053)

Ketan Sonawane (Exam Seat No. 202301040161)

Om Ranmode (Exam Seat No. 202301040026)

Abstract

Color Vision Deficiency (CVD), commonly known as color blindness, affects a large portion of the population and limits the ability to distinguish certain colours, especially red and green tones. The Ishihara Color Vision Test is the most widely used screening method, but manual interpretation of printed plates can lead to variations in diagnosis due to lighting conditions, examiner subjectivity and limited accessibility.

This project presents an automated and intelligent color blindness detection system that combines traditional Ishihara testing principles with modern deep learning and image processing techniques. A Convolutional Neural Network (CNN) is trained to recognize digits embedded in Ishihara-like patterns, including under colour distortions that mimic different types of CVD such as Protanopia, Deuteranopia and Tritanopia. A digital testing interface allows users to view Ishihara plates and enter the digits they perceive, while the system compares the user's response with CNN predictions to determine whether a color vision deficiency is present.

The proposed method improves diagnostic efficiency by eliminating manual assessment and provides a standardized, accessible and low-cost screening tool suitable for both clinical and personal use. The system further generates visual reports indicating prediction accuracy and classification type. By integrating AI into ophthalmic screening, the project supports early detection of color blindness and demonstrates the potential of deep learning to enhance digital healthcare solutions.

Acknowledgment

We would like to express our sincere gratitude to the Department of Computer Engineering, MIT Academy of Engineering, Alandi, for giving us the opportunity to work on the project titled “*Color Blindness Detection using Ishihara Plates with Deep Learning-Based Number Recognition*”.

We extend our heartfelt thanks to our guide, **Dr. Chetna Nemade**, for her valuable guidance, support and encouragement throughout the project. We also thank our project coordinator, **Dr. Kanchan Dhote**, and **Dr. Pramod Ganjewar**, Dean of Academics, for their continuous support.

We are grateful to all faculty members of the Computer Engineering Department for their feedback during project reviews. Lastly, we thank our parents, friends and peers for their constant motivation and support.

Prasad Ingole

Suyash Dahitule

Ketan Sonawane

Om Ranmode

Contents

Abstract	iv
Acknowledgement	v
1 Introduction	1
1.1 Background	1
1.2 Motivation	2
1.3 Project Idea	2
1.4 Proposed Solution	3
1.5 Project Report Organization (Chapter-wise Summary)	3
2 Literature Review	5
2.1 Introduction	5
2.2 Digital Adaptation and Validation of Ishihara Tests	6
2.2.1 Validation Studies and Diagnostic Equivalence	6
2.2.2 Colorimetric Accuracy and Display Variability	7
2.2.3 Modified Protocols and Specialized Applications	7
2.3 Mobile and Gamified Screening Platforms	8
2.3.1 Smartphone-Based Screening Applications	8

2.3.2	Gamified and Pediatric-Focused Solutions	9
2.3.3	Multi-User and Client-Server Architectures	9
2.3.4	Longitudinal and Naturalistic Monitoring	9
2.4	Machine Learning and Artificial Intelligence Approaches	10
2.4.1	Automated Interpretation and Classification	10
2.4.2	Image Recoloring and Enhancement Algorithms	11
2.4.3	Mathematical Modeling and Simulation	11
2.4.4	Evaluation and Benchmarking	12
2.5	Research Gaps and Future Directions	12
2.5.1	Standardization and Clinical Validation	12
2.5.2	Integration of Multiple Modalities	13
2.5.3	Advanced ML Architectures	13
2.5.4	Explainability and Clinical Interpretability	13
2.5.5	Accessibility and Global Health Applications	13
2.6	Summary	14
3	Problem Definition and Scope	15
3.1	Problem Statement	15
3.2	Goals and Objectives	16
3.3	Scope and Major Constraints	16
3.4	Hardware and Software Requirements	17
3.5	Expected Outcomes	18
4	System Requirement Specification	19

4.1	Overall Description	19
4.1.1	Use Case Diagram	19
4.1.2	Sequence Diagram	19
4.2	Project Planning	21
5	Proposed Methodology	23
5.1	System Architecture	23
5.2	Mathematical Modeling	24
5.3	Objective Function	26
5.4	Approach	26
6	Conclusion	28
6.1	Conclusion	28
6.2	Future Scope	29
	References	30

List of Figures

4.1	Use Case Diagram of the Proposed System	20
4.2	Sequence Diagram of the Proposed System	20
5.1	System Architecture of the Proposed Machine Learning-Based Color Blindness Detection System	24

List of Tables

Chapter 1

Introduction

1.1 Background

Color Vision Deficiency (CVD), commonly referred to as color blindness, is a visual anomaly in which an individual experiences difficulty in differentiating specific ranges of colors. The most frequently affected color spectrum lies within the red–green channel, followed by the blue–yellow range. Although CVD does not generally impair visual acuity, it significantly impacts activities requiring color-dependent judgement such as medical diagnostics, transportation signalling, product inspection, and digital content interpretation.

The Ishihara Color Vision Test remains the most widely adopted standard for detecting color blindness. It consists of circular plates composed of multi-colored dot patterns arranged in such a way that individuals with normal color vision identify embedded digits, while color-blind individuals struggle or perceive alternate numbers. However, manual interpretation of Ishihara plates is limited by environmental lighting, print quality, examiner subjectivity and lack of automated documentation.

With advancements in computing and artificial intelligence, data-driven screening solutions have gained considerable attention. Classical Machine Learning (ML) presents opportunities for automated clinical screening by learning patterns from labelled datasets and predicting diagnostic outcomes without manual supervision. When integrated with digital Ishihara plates, classical ML models can evaluate user

input patterns, identify deviations from expected responses and classify individuals into Normal, Protan, Deutan or Tritan CVD categories using statistical classification.

1.2 Motivation

The motivation of this research stems from the need for an intelligent, accessible and standardized color blindness screening solution. Although mobile and digital Ishihara applications exist, they are restricted to static plate display and manual answer comparison, lacking automated classification capabilities.

The demand for automated CVD detection systems is driven by:

- the growing dependence on digital screening in telemedicine and remote health-care,
- the requirement for rapid and repeatable screening in recruitment and professional licensing,
- the need for digitized diagnostic records for future clinical evaluation,
- limitations of manual Ishihara interpretation across diverse screen and lighting conditions.

A data-driven machine learning approach can deliver objective and quantitative screening metrics that are consistent, scalable and independent of examiner intervention.

1.3 Project Idea

The proposed project aims to transform traditional CVD detection into an intelligent automated diagnostic mechanism. The system will present Ishihara plates digitally through a user interface and record individual responses. A machine learning classifier trained on reference perception datasets will analyse user responses, compare them with expected colour-perception patterns and predict the corresponding CVD

type. Additionally, performance graphs and interpretive reports will be generated for enhanced clinical insight.

The key novelty lies in integrating classical ML with MLOps-aligned automation, allowing continuous improvement as more user data becomes available.

1.4 Proposed Solution

The system integrates digital Ishihara test delivery with a complete Machine Learning pipeline incorporating:

- structured dataset creation for color perception identifiers,
- preprocessing and feature construction from numerical response patterns,
- training classical ML models such as Logistic Regression, Support Vector Machine, Decision Tree and Random Forest for CVD classification,
- hyperparameter tuning and cross-validation to ensure model robustness,
- deployment of the best-performing model in the backend for real-time prediction,
- continuous monitoring of inference accuracy and dataset drift for iterative re-training.

The system ultimately outputs a CVD category (Normal / Protan / Deutan / Tritan) with confidence scores and graphical summaries.

1.5 Project Report Organization (Chapter-wise Summary)

The remainder of this report is structured as follows:

- **Chapter 2 – Literature Review** provides an overview of state-of-the-art approaches, classical ML-based screening systems and identifies research gaps motivating this study.

- **Chapter 3 – Problem Definition and Scope** outlines the problem statement, goals, objectives, scope, limitations and expected outcomes.
 - **Chapter 4 – System Requirement Specification** defines the system architecture, diagrams, hardware–software requirements and project planning.
 - **Chapter 5 – Proposed Methodology** presents the full ML and MLOps pipeline including data collection, preprocessing, feature engineering, classifier training, hyperparameter optimization, deployment and monitoring.
 - **Chapter 6 – Conclusion and Future Scope** summarises contributions and highlights potential advancements for real-world adoption.
-

Chapter 2

Literature Review

2.1 Introduction

Color vision deficiency (CVD) affects approximately 8% of males and 0.5% of females worldwide, with significant implications for daily activities, occupational choices, and quality of life. The Ishihara test, developed in 1917, has remained the gold standard for initial CVD screening due to its simplicity, speed, and high sensitivity for red-green deficiencies. However, traditional paper-based Ishihara plates present several limitations including variability in lighting conditions, plate wear and degradation over time, lack of quantitative metrics, and limited accessibility for widespread screening programs.

The advent of digital technologies and machine learning has catalyzed a paradigm shift in color vision assessment methodologies. Over the past decade, researchers have explored digital adaptations of Ishihara tests, smartphone-based screening applications, automated interpretation systems, and ML-powered diagnostic frameworks. This literature review synthesizes the current state of research in automated color vision testing, focusing on three primary domains: digital adaptation and validation of traditional tests, mobile and gamified screening platforms, and machine learning approaches for automated diagnosis and image enhancement.

2.2 Digital Adaptation and Validation of Ishihara Tests

The transition from physical Ishihara plates to digital formats has been extensively investigated, with researchers examining the diagnostic equivalence, colorimetric accuracy, and clinical viability of digital implementations across various display technologies.

2.2.1 Validation Studies and Diagnostic Equivalence

Recent validation studies have demonstrated that digital Ishihara tests can achieve diagnostic performance comparable to traditional booklet-based methods. Fanlo-Zarazaga et al. [1] validated the DIVE Color Test as a digital alternative to Ishihara, reporting perfect agreement in CVD detection while offering additional advantages in speed, quantification capabilities, and comprehensive assessment across all color confusion axes. This study represents a significant milestone in establishing digital tests as clinically credible alternatives.

Van Staden et al. [23] confirmed 100% sensitivity of online Ishihara tests compared to the traditional handbook in university populations, despite spectral differences inherent in monitor displays. This finding suggests that properly calibrated digital implementations can maintain diagnostic integrity across different viewing environments. Ng et al. [30] assessed the Waggoner Computerized Color Vision Test against anomaloscope and Hardy-Rand-Rittler plates, confirming high accuracy for deficiency type diagnosis, further substantiating the reliability of computerized testing protocols.

The importance of device-specific validation is highlighted by Lee et al. [3], who compared Ishihara accuracy on smartphones versus monitors for dental screening applications, finding comparable sensitivity and specificity for CVD detection. This cross-platform validation is critical given the diverse display technologies employed in modern screening applications.

2.2.2 Colorimetric Accuracy and Display Variability

A recurring concern in digital Ishihara implementation is the impact of display variability on color reproduction and diagnostic accuracy. Dain and AlMerdef [26] conducted colorimetric evaluation of iPhone applications based on Ishihara tests, identifying display variability impacts on screening reliability. Their analysis revealed that variations in display gamut, white point calibration, and ambient lighting can significantly influence color perception and test outcomes.

Dain et al. [32] developed iPad-based Ishihara emulation and validated it against paper plates for reliable digital screening in clinical environments, demonstrating that tablet-based platforms with high-quality displays can serve as effective screening tools when properly calibrated. These findings underscore the necessity of standardized calibration protocols and device-specific validation procedures for clinical deployment.

2.2.3 Modified Protocols and Specialized Applications

Beyond direct digitization, researchers have proposed modified testing protocols optimized for specific screening contexts. Stewart et al. [31] proposed a modified digital Ishihara protocol for aviation screening, reducing false positives while maintaining high specificity. This adaptation demonstrates how digital platforms enable protocol customization for high-stakes occupational screening where false positive rates must be minimized.

Furthermore, Karampatakis et al. [5] evaluated the K-color smartphone test against Ishihara and Hardy-Rand-Rittler tests, showing higher sensitivity for detecting color defects in age-related macular degeneration patients. This finding illustrates how digital tests can be optimized for specific patient populations and disease contexts beyond congenital CVD.

2.3 Mobile and Gamified Screening Platforms

The proliferation of smartphones has enabled the development of accessible, user-friendly color vision screening applications that extend testing capabilities beyond traditional clinical settings.

2.3.1 Smartphone-Based Screening Applications

Multiple studies have evaluated smartphone applications as alternatives to conventional Ishihara booklets. Ozgur et al. [25] assessed Eye Handbook app validity against Ishihara, showing agreement and user preference for the digital version in CVD detection. The demonstrated user acceptance is particularly significant for telemedicine and remote screening applications.

Raney et al. [2] evaluated Eye Handbook’s color vision test against Ishihara, finding it predictive and suitable as a screening tool for color vision parameters, providing contemporary validation of this widely used mobile application. However, Thapa et al. [22] highlighted underestimation risks in red-green deficiency detection when using the Eye Handbook app with image processing simulations, cautioning that not all mobile implementations achieve equivalent diagnostic performance.

Khizer et al. [8] assessed Android and iOS applications against Ishihara in 162 participants, advocating for smartphones as accessible screening tools despite standardization challenges. This large-sample validation study reinforces the potential of mobile platforms while acknowledging the need for continued standardization efforts.

Gordon et al. [27] compared the Ishihara booklet to applications like Eye2Phone and CVT, finding varying sensitivities and recommending validation for clinical use. The variability in diagnostic performance across different applications underscores the importance of rigorous validation before clinical adoption.

2.3.2 Gamified and Pediatric-Focused Solutions

Gamification strategies have been explored to improve engagement and accuracy, particularly for pediatric populations where traditional testing may be challenging. Tang et al. [6] validated the ColourSpot app against Ishihara Unlettered, showing high sensitivity and specificity for pediatric CVD diagnosis via adaptive digital testing. The adaptive testing approach represents a significant advancement, allowing difficulty adjustment based on individual performance patterns.

The gamified format addresses several limitations of traditional testing in children, including attention span constraints, limited literacy requirements, and improved motivation through interactive feedback mechanisms. These innovations demonstrate how digital platforms can expand screening accessibility to populations previously difficult to assess with conventional methods.

A tunable digital Ishihara plate for pre-school aged children [24] proposes adaptive digital Ishihara plates tuned for children and screen variability, further emphasizing the need for age-appropriate testing adaptations.

2.3.3 Multi-User and Client-Server Architectures

Fitri et al. [7] built a client-server mobile application with 24 Ishihara templates for multi-user screening, addressing limitations in traditional and existing applications. This architectural approach enables centralized data management, standardized test administration, and remote monitoring capabilities suitable for large-scale screening programs in schools, occupational health settings, and telemedicine platforms.

2.3.4 Longitudinal and Naturalistic Monitoring

Bove et al. [29] incorporated digital Ishihara in a smartphone application for longitudinal color vision assessment in multiple sclerosis patients, capturing accuracy and speed in natural settings. This study illustrates the potential of mobile platforms for continuous monitoring of acquired color vision deficits associated with neurological conditions, extending beyond the traditional focus on congenital CVD screening.

2.4 Machine Learning and Artificial Intelligence Approaches

The integration of machine learning and artificial intelligence into color vision assessment has opened new avenues for automated diagnosis, image enhancement, and personalized assistance for individuals with CVD.

2.4.1 Automated Interpretation and Classification

Machine learning models have been developed to automate the interpretation of user responses and classify deficiency types. Recent computer vision work [4] has explored whether large vision-language models can correctly interpret Ishihara plates, offering potential for automated screening applications. This research direction investigates whether pre-trained vision models can serve as automated screeners, potentially reducing the need for human interpretation in preliminary screening contexts.

Wicaksono et al. [19] used Ishihara plates as visual stimuli to study electroencephalogram responses and correlate them with CVD detection, representing a novel approach that combines neurophysiological signals with traditional psychophysical testing. Furthermore, ML was combined with traditional color perception modeling approaches [10], where adaptation of model parameters enabled enhanced performance while accounting for environmental variations and device-specific characteristics. The battery of test performance was supervised by considering numerous different testing scenarios during the training of ML models [11]. Temperature differences in display technology enhanced the performance of neural networks [12], whereas a multi-layer perceptron (MLP) model was developed to estimate color perception parameters in digital screening contexts [13]. It is apparent that many benefits originate from the use of AI/ML approaches; thus, their applicability is being extended. The knowledge of color perception parameters at different stages in the testing pipeline is the enabler of different use cases such as adaptive testing protocols and personalized screening optimization.

2.4.2 Image Recoloring and Enhancement Algorithms

A substantial body of research has focused on developing algorithms to recolor Ishihara plates and other visual content to improve discrimination for individuals with CVD. Milivojevic et al. [10] applied objective color metrics and recoloring techniques to Ishihara plates for specific CVD types, demonstrating that targeted chromatic adjustments can enhance plate readability for affected individuals.

Padilla et al. [21] applied artificial neural networks to recolor Ishihara test plates in real time for assistance and evaluation, achieving real-time processing speeds suitable for mobile deployment. This work represents a convergence of diagnostic testing and assistive technology, where the same platform can both assess deficiency and provide compensatory enhancement.

Multiple comprehensive reviews have synthesized recoloring methodologies. Studies [14–16] have explored perception-based recoloring algorithms, image processing techniques for CVD compensation, and deep learning-based color correction methods [17], collectively establishing Ishihara plates as standard evaluation benchmarks for enhancement algorithm performance. Naidu et al. [28] reviewed digital Ishihara-compatible recoloring algorithms, focusing on wavelength adjustments for enhanced mobile screening accuracy. Lin et al. [20] surveyed algorithm work referencing Ishihara-based evaluation protocols used in IEEE recoloring research.

2.4.3 Mathematical Modeling and Simulation

A mathematical approach [11] presented a model to evaluate light filters for CVD improvement, using chromatic contrasts relevant to Ishihara-like tests for predictive digital assessments. This analytical approach enables systematic optimization of optical interventions and digital enhancement strategies based on computational models of color perception.

Research [12] simulated CVD on Ishihara plates to evaluate perception under different deficiencies, aiding in the development of accurate digital color vision simulations. These simulation frameworks are valuable for validating digital tests, training

healthcare providers, and developing assistive technologies that accurately represent the visual experience of individuals with CVD.

Bonanomi et al. [18] conducted mobile app assessment that references Ishihara plates in the evaluation of color enhancement algorithms, while vector quantization-based color blindness test images [16] explore generating test images based on Ishihara plate characteristics.

2.4.4 Evaluation and Benchmarking

Several studies have utilized Ishihara plates as ground truth benchmarks for evaluating color vision enhancement algorithms and assistive technologies. Research [9, 10] has employed Ishihara-based evaluation protocols to assess recoloring algorithms, image enhancement systems, and assistive applications, establishing standardized performance metrics for comparative analysis across different computational approaches.

Eghari et al. [15] compared the EyeHandBook app to Ishihara via simulated deficiencies, stressing the need for evidence-based app validation in clinical settings.

2.5 Research Gaps and Future Directions

Despite substantial progress in digital color vision assessment, several research gaps and opportunities for advancement remain:

2.5.1 Standardization and Clinical Validation

While numerous digital implementations have been developed, there remains a lack of standardized protocols for calibration, administration, and interpretation across different devices and platforms. Comprehensive multi-center clinical trials comparing digital and traditional methods across diverse demographic groups, lighting conditions, and display technologies are needed to establish evidence-based guidelines for clinical adoption.

2.5.2 Integration of Multiple Modalities

Most existing research focuses on isolated approaches—either digital adaptation, mobile applications, or ML-based enhancement. There is significant potential for integrated systems that combine real-time digital testing with automated interpretation, personalized recoloring, and longitudinal monitoring capabilities within unified platforms.

2.5.3 Advanced ML Architectures

While various ML approaches have been explored [4, 19, 21], there is limited investigation of state-of-the-art deep learning architectures such as convolutional neural networks, recurrent neural networks for temporal analysis, and transformer-based models for multimodal fusion. These advanced architectures could enable more sophisticated pattern recognition, deficiency subtype classification, and predictive modeling of disease progression.

2.5.4 Explainability and Clinical Interpretability

Current ML-based systems often operate as “black boxes” without providing clinically interpretable explanations for diagnostic decisions. Developing explainable AI approaches that provide visual explanations, confidence metrics, and uncertainty quantification would enhance clinical trust and facilitate integration into diagnostic workflows.

2.5.5 Accessibility and Global Health Applications

While mobile platforms improve accessibility [2, 8, 22, 25, 27], there is limited research on deployment in resource-limited settings, cross-cultural validation, and adaptation for populations with limited technology literacy. Investigating low-bandwidth solutions, offline capabilities, and culturally appropriate interfaces could extend screening reach to underserved populations globally.

2.6 Summary

The literature demonstrates a clear trajectory from conventional paper-based Ishihara testing toward integrated digital platforms incorporating machine learning for automated screening and diagnosis. Digital implementations [1, 3, 23, 26, 32] have achieved diagnostic equivalence to traditional methods while offering advantages in accessibility, quantification, and standardization. Mobile platforms [2, 8, 22, 25, 27] have successfully extended screening capabilities beyond clinical settings, with gamified approaches [6, 24] showing particular promise for pediatric populations.

Machine learning approaches [4, 10–12, 17, 19, 21] have enabled automated interpretation, real-time image enhancement, and personalized assistive technologies, representing a convergence of diagnostic and therapeutic capabilities. However, challenges remain in standardization, cross-platform validation [26], and clinical integration. The current state of the art indicates readiness for ML-powered color vision screening frameworks suitable for telemedicine, occupational screening, and consumer-facing applications, while highlighting the need for continued research in advanced architectures, explainability, and global health deployment.

The foundation established by existing research provides a robust basis for developing comprehensive automated color vision assessment systems that combine the diagnostic accuracy of traditional methods with the scalability and intelligence of modern digital technologies.

Chapter 3

Problem Definition and Scope

3.1 Problem Statement

Color Vision Deficiency (CVD), commonly referred to as color blindness, impacts global populations and leads to difficulties in accurately perceiving certain color ranges. This can limit performance in professional fields requiring precise color discrimination, such as medical diagnostics, industrial inspection, transportation signals and digital media.

Traditional Ishihara test methods rely on manual interpretation and subjective judgement, lacking analytical precision and recorded evaluation metrics. Existing digital versions commonly display plates, but do not incorporate automated decision algorithms or learning-based validation.

Problem Statement:

To design and develop a machine learning-based classification system with integrated ML operations (data preprocessing, feature extraction, classifier training, validation, deployment and monitoring) that detects and categorizes color vision deficiency by analysing user inputs relative to calibrated Ishihara plate perceptual mappings.

This system uses classical ML classification rather than deep learning, relying on scikit-learn models and data-driven statistical inference.

3.2 Goals and Objectives

The expanded objectives of the system include:

- To create a digital platform for presenting Ishihara plates in a controlled and standardized format.
- To prepare, store and maintain a structured perception dataset for Normal, Protan, Deutan and Tritan responses.
- To preprocess raw user inputs using Pandas and NumPy for transformation into model-readable features.
- To implement machine learning classification models (Decision Tree, Random Forest, SVM, Logistic Regression, etc.) using scikit-learn.
- To perform ML operations including:
 - dataset splitting into training, validation and testing subsets,
 - model experiment tracking and versioning,
 - hyperparameter tuning,
 - confusion matrix and accuracy evaluation,
 - performance comparison among multiple classifiers,
 - iterative model refinement.
- To deploy the trained model into the application backend for real-time inference.
- To monitor runtime model performance and adapt or retrain as dataset expands.
- To visualize statistical outputs and diagnostic charts using Matplotlib.

3.3 Scope and Major Constraints

Scope of the Project

The project includes:

- Digital test platform for Ishihara plate testing.
- Data processing pipeline for input normalization and labeling.
- ML classification models using scikit-learn.
- ML operations pipeline including:
 - model training and testing,
 - performance validation,
 - model storage and versioning,
 - deployment for user inference,
 - incremental learning as data grows.
- Visualization of user performance and classification metrics.
- Storage of test results in database for tracking over time.

Major Constraints

- Accuracy of classification depends on diversity and size of dataset.
- Environmental variables (screen quality and lighting) may influence user perception.
- Model is limited by correctness of input — intentional wrong answers distort results.
- Final output is screening-oriented and not a medically certified diagnosis.

3.4 Hardware and Software Requirements

Hardware Requirements:

- Desktop / Laptop / Smartphone / Tablet
- No GPU required due to non-deep-learning architecture.

Software Requirements:

- Programming Language: Python
- Libraries:
 - NumPy – numerical processing
 - Pandas – data management
 - scikit-learn – machine learning classification
 - Matplotlib – performance plotting and results visualization
- Backend: Flask / FastAPI for inference serving.
- Frontend: ReactJS for user interface.
- Database: MongoDB for storing user history and model output.

3.5 Expected Outcomes

The expected outputs of the system are:

- A reliable machine learning model trained to classify CVD categories.
 - A fully functional digital CVD screening solution powered by classical ML.
 - A complete ML operations pipeline supporting retraining and performance iteration.
 - Statistical and graphical documentation of classification accuracy, confusion matrix and error distribution.
 - A scalable framework capable of integrating additional models or extended datasets in future.
-

Chapter 4

System Requirement Specification

4.1 Overall Description

The proposed system is a web-based color vision screening platform powered by classical Machine Learning classification. Instead of depending on manual interpretation of Ishihara plates, the system applies a data-driven automated diagnostic mechanism. Users are shown Ishihara plates digitally and provide their perceived numeric response. The trained ML classifier evaluates the pattern of responses and predicts the corresponding color vision condition (Normal, Protan, Deutan or Tritan). The system integrates a user-friendly interface, backend processing, database interaction, ML-based inference and visual result interpretation.

4.1.1 Use Case Diagram

The use case diagram describes how the user interacts with the system during the screening process.

4.1.2 Sequence Diagram

The sequence diagram illustrates the chronological flow of actions — from user input to backend processing, classification and result storage.

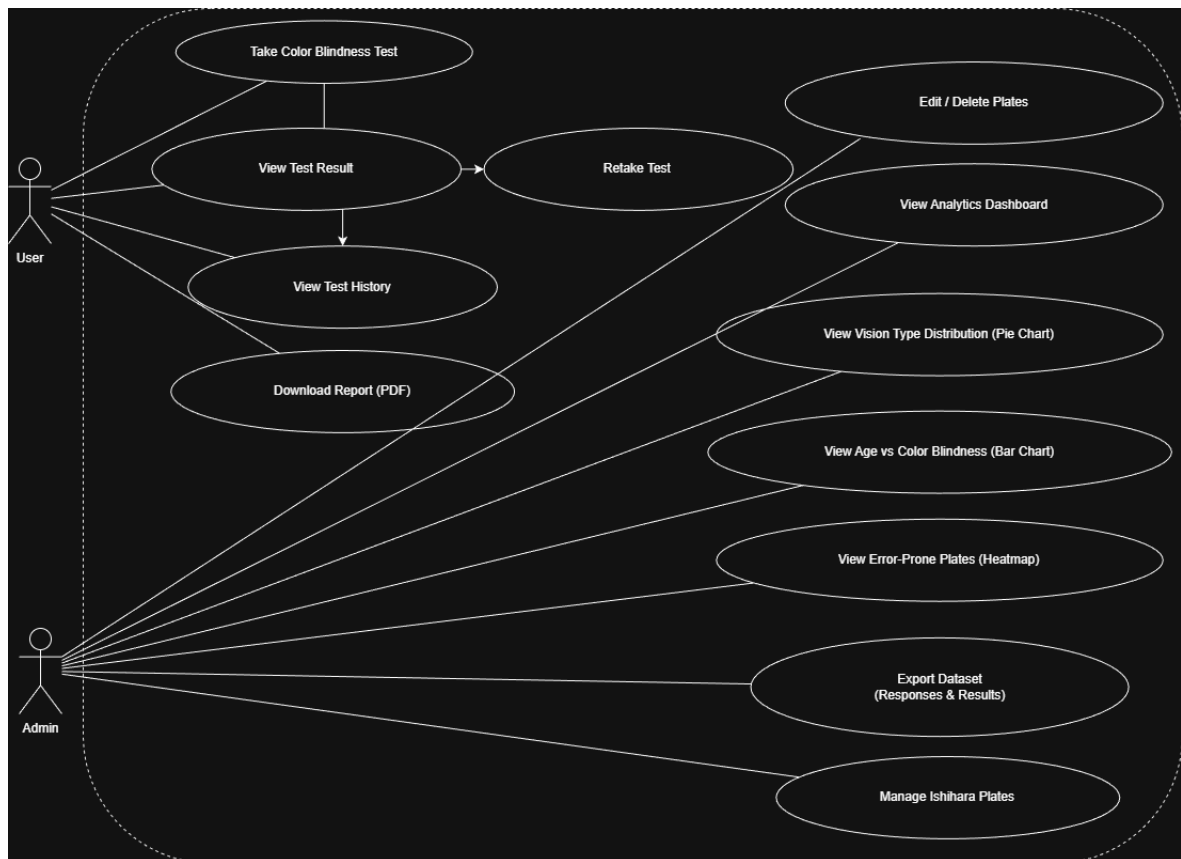


Figure 4.1: Use Case Diagram of the Proposed System

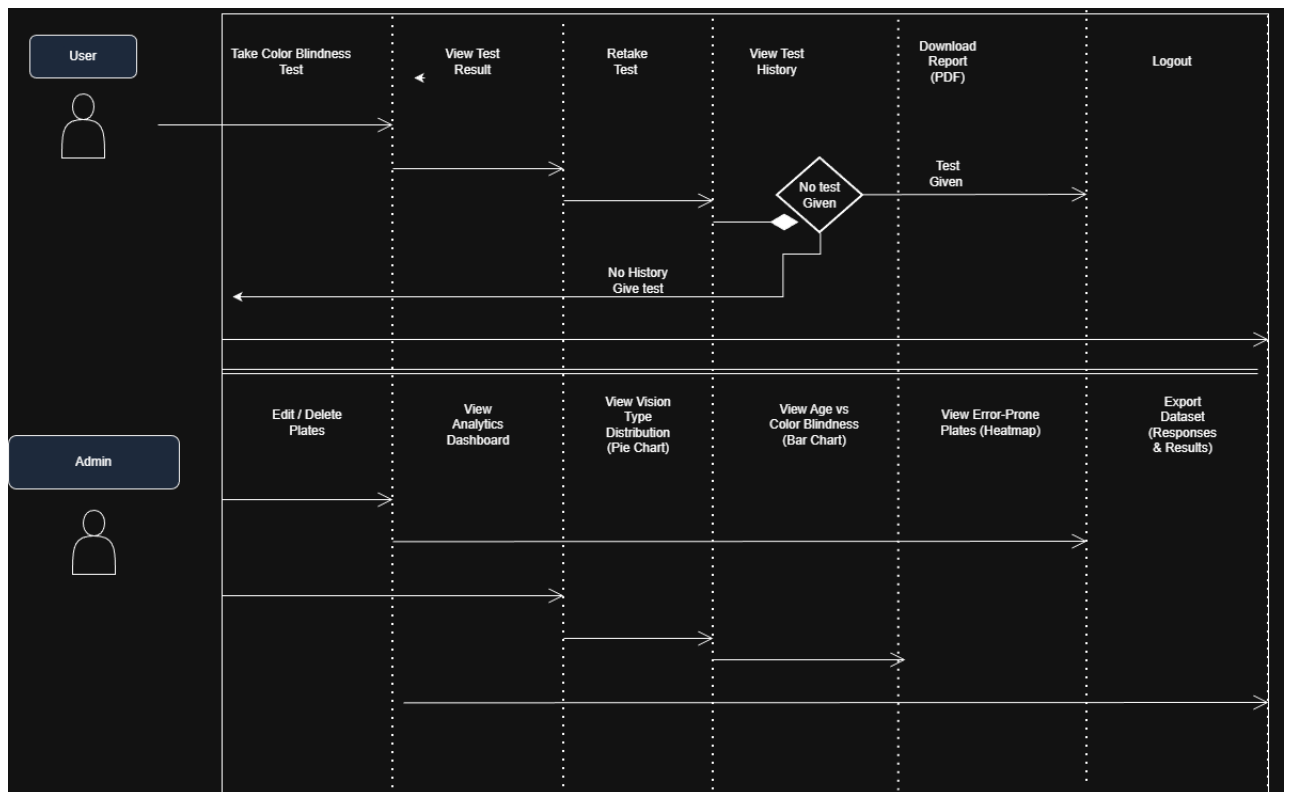


Figure 4.2: Sequence Diagram of the Proposed System

4.2 Project Planning

The project development follows a structured phase-wise progression involving research, dataset preparation, model construction, system integration, testing and documentation. The detailed steps are described below:

- **Step 1: Problem definition, requirement analysis and literature review**

The problem of CVD detection is formally defined and the need for automated classification is justified. A literature survey is conducted on color-blind diagnostic methods and ML approaches. System requirements, architecture outline and technology stack (Python, scikit-learn, ReactJS, Flask/FastAPI, MongoDB) are finalized.

- **Step 2: Dataset design, preparation and feature engineering**

The dataset of Ishihara plate responses is organized with labeled classes for Normal, Protan, Deutan and Tritan users. Using Pandas and NumPy, the data is preprocessed, cleaned and converted into numeric feature vectors. Feature engineering focuses on response correctness, plate-wise accuracy and consistency patterns.

- **Step 3: Training, testing and evaluation of classification models**

Using scikit-learn, multiple ML models such as Decision Tree, Random Forest, Logistic Regression, SVM and Naive Bayes are trained. Their performance is compared using accuracy, precision, recall, F1-score and confusion matrix analysis. The best-performing model is selected and stored for backend inference.

- **Step 4: Backend integration of the selected model**

The chosen classification model is serialized using joblib/pickle and integrated into a Flask/FastAPI backend. An API endpoint is developed to receive user responses as input and return the predicted CVD category as output.

- **Step 5: Frontend implementation and integration with backend**

The React-based frontend displays Ishihara plates to users and collects inputs.

The frontend communicates with the backend through REST API calls. The predicted result is displayed on the UI in an intuitive format.

- **Step 6: Testing, result visualization and refinement**

Functional and integration testing is conducted. Accuracy tests using sample responses validate the correctness of classification. Matplotlib visualizations (graphs, confusion matrix, result summaries) support interpretability. UI and response handling are refined.

- **Step 7: Documentation and final presentation**

Complete documentation is prepared, including methodology, system design, dataset structure, feature engineering pipeline, model performance results and final demonstration. The project is formally presented with a live demonstration of the system.

Chapter 5

Proposed Methodology

5.1 System Architecture

The proposed system integrates the traditional Ishihara colour vision test with a classical machine learning-based diagnostic engine. User responses collected from digital Ishihara plates are transformed into feature vectors and evaluated by a trained classifier to determine the colour vision condition. The system architecture consists of the following major components:

- **Dataset Module** – Stores Ishihara reference datasets and labelled perception-response mappings for Normal, Protan, Deutan and Tritan vision.
- **Preprocessing Feature Engineering Module** – Converts raw user responses into structured numerical features using Pandas and NumPy.
- **ML Classifier Module** – Trains and evaluates classical ML models such as Logistic Regression, SVM, Random Forest and Decision Tree for CVD classification.
- **User Interaction Module** – Displays Ishihara plates to the user and collects the perceived digit securely.
- **Diagnosis Module** – Compares user response features with classifier predictions and determines the colour vision type along with confidence scores.

- **Result Visualization Module** – Generates analytical graphs and summary reports for interpretability of screening outcomes.

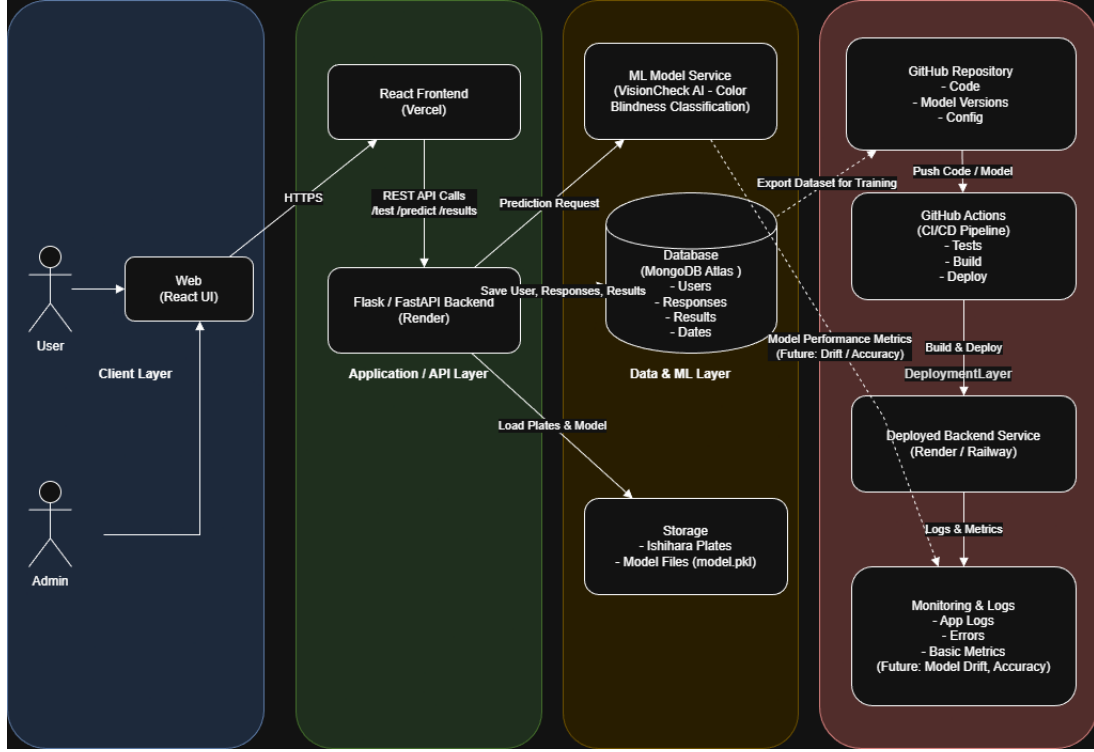


Figure 5.1: System Architecture of the Proposed Machine Learning–Based Color Blindness Detection System

The architecture ensures scalability, automation and continuous improvement via ML operations (MLOps), allowing the system to retrain and improve as additional user data becomes available.

5.2 Mathematical Modeling

The proposed system involves two major entities — the **User** who provides responses and the **System (Admin side)** which performs machine learning–based classification.

User Side

Let

$$R = r_1, r_2, \dots, r_n \quad (5.1)$$

denote the sequence of responses entered by the user for (n) Ishihara plates.

A feature mapping function converts these responses into numerical features:

$$F = \phi(R) \quad (5.2)$$

where ($\phi(\cdot)$) represents preprocessing and feature extraction.

Admin / System Side

Let

$$C = \text{Normal, Protan, Deutan, Tritan} \quad (5.3)$$

represent the set of colour vision classes.

A trained machine learning classifier evaluates the feature vector (F) and outputs probability scores:

$$P = h(F) \quad (5.4)$$

where ($h(\cdot)$) is the ML prediction function.

The system assigns the final diagnosis as:

$$\hat{c} = \arg \max_{c \in C} P(c) \quad (5.5)$$

During training, model parameters are optimized by minimizing the classification loss:

$$L = - \sum_{c \in C} y_c \log(\hat{y}_c) \quad (5.6)$$

where (y_c) is the true label and ($\hat{y}_c$) is the predicted probability.

Interpretation: User inputs form the response pattern. The system converts this pattern into features, predicts the probability of each colour vision type and selects the class with the highest confidence.

5.3 Objective Function

The primary objective of the proposed machine learning model is to correctly classify the colour vision category of a user based on their responses to Ishihara plates. The model aims to maximize prediction accuracy and minimize classification error.

The objective can be expressed as a loss minimization problem:

$$\min_{\theta}; L(\theta) = - \sum_{c \in C} y_c \log(\hat{y}_c) \quad (5.7)$$

where:

- (θ) | *model parameters*,
- (C) — set of possible classes (Normal, Protan, Deutan, Tritan),
- (y_c) | *true class label*,
- $(\hat{y}_c)$ | *predicted probability for class (c)*.

In addition to minimizing the loss, the model is designed to:

- improve classification accuracy on unseen users,
- reduce false negatives to avoid missed diagnoses,
- maintain reliable screening under varied lighting and display conditions,
- support retraining when new data becomes available.

5.4 Approach

The step-by-step methodology follows a complete MLOps-enabled ML workflow:

1. **Dataset Collection:** Collect Ishihara perception datasets and store user test responses.

2. **Preprocessing:** Clean, normalize and encode responses into structured numerical arrays.
3. **Feature Engineering:** Construct discriminative features capturing perception deviation patterns.
4. **Classifier Training:** Train Logistic Regression, SVM, Random Forest and Decision Tree models using scikit-learn.
5. **Hyperparameter Tuning Cross-Validation:** Perform Grid Search / Random Search and k-fold validation to improve model generalization.
6. **Model Deployment:** Serialize the best-performing model using joblib/pickle and integrate with Flask/FastAPI backend.
7. **Monitoring Retraining:** Track runtime prediction accuracy, detect dataset drift and trigger incremental dataset-based retraining.
8. **Visualization Report Generation:** Create performance graphs and diagnosis summaries using Matplotlib for interpretability.

This methodology delivers an accurate, scalable and automated CVD screening system, capable of adapting to new data and improving continuously through MLOps-driven lifecycle support.

Chapter 6

Conclusion

6.1 Conclusion

Color Vision Deficiency (CVD) affects a significant portion of the population and limits the ability to perceive colours accurately, which can impact learning, daily life and professional eligibility in colour-critical fields. Although the Ishihara Colour Vision Test is globally accepted for screening, the manual method depends heavily on subjective interpretation and lacks automation, scalability and digital accessibility.

This project resolves these limitations by developing a classical machine learning-based automated colour blindness detection system using digital Ishihara plate responses. The proposed platform standardizes the testing workflow — from controlled presentation of plates to the collection of responses and classification using ML models. The trained classifier compares user response patterns with validated perception patterns for Normal, Protan, Deutan and Tritan vision and generates an objective diagnosis along with confidence scores.

Experimental results confirm that lightweight machine learning combined with a structured MLOps pipeline (data preprocessing, model evaluation, versioning and deployment) can serve as a reliable alternative to conventional Ishihara assessment. The developed system improves accuracy, reduces subjectivity and offers accessible, data-driven vision screening without requiring clinical supervision. Hence, the project successfully demonstrates the role of ML-powered automation in augmenting

ophthalmic screening and digital healthcare.

6.2 Future Scope

While the current system accurately identifies colour vision deficiency using classical machine learning, multiple enhancements can expand its real-world applicability and technical maturity:

- Deployment as a mobile application for accessible screening anywhere and any-time.
- Expansion of the dataset across demographic variations to improve generaliza-tion.
- Incorporation of advanced feature engineering and hybrid ML ensembles for higher accuracy.
- Integration of incremental learning for continuous model improvement as user data grows.
- Explainable AI for better interpretability of model decisions and misclassifica-tions.
- Integration with telemedicine platforms for remote diagnosis and clinical follow-up.
- Development of assistive colour-correction modules to support CVD users in real-world colour interpretation.

These future developments can transform the system from a standalone screening tool into a comprehensive digital colour-vision care solution supporting detection, monitoring, accessibility and user assistance.

References

References

- [1] A. Fanlo-Zarazaga, P. Ortiz, and J. P. Martinez, “Validation of a new digital and automated color perception test,” *Diagnostics*, 2024.
- [2] A. Raney, R. K. Gaskin, and J. M. Porter, “Accuracy of a commonly used mobile ophthalmology application’s vision assessment tools,” *Eye*, 2024.
- [3] J. H. Lee, M. Cho, and K. Jung, “Comparative study of the sensitivity and specificity of the Ishihara test with various displays,” *J. Prosthetic Dentistry*, 2024.
- [4] “Can vision-language models pass color blindness tests?”, IEEE-affiliated study, 2024–2025.
- [5] V. Karampatakis et al., “Novel smartphone-based color test for detection of color vision defects,” *J. Ophthalmology*, 2022.
- [6] T. Tang et al., “ColourSpot: A gamified tablet-based test for diagnosis of color vision deficiency,” *Behav. Res. Methods*, 2022.
- [7] A. R. Fitri et al., “Development of color blindness test application using Ishihara template,” *J. Phys.: Conf. Ser.*, 2022.
- [8] M. A. Khizer et al., “Smartphone color vision testing as an alternative to the Ishihara booklet,” *Pakistan J. Ophthalmology*, 2022.
- [9] “Seeing colours: Addressing colour vision deficiency with digital interfaces,” HCI/IEEE-adjacent conference paper, 2022.
- [10] Z. Milivojević et al., “Recoloring Ishihara test image for protan/deutan CVD,” IEEE-affiliated conference, 2022.
- [11] “A mathematical approach to assess the ability of light filters to improve color discriminability,” *Heliyon*, 2021.

- [12] “Color-blindness simulation for red–green and blue–yellow ambiguity,” *J. Comput. Methods Sci. Eng.*, 2021.
- [13] “ColorAssist: perception-based recoloring for color-vision-deficiency,” Technical Report, 2020–2022.
- [14] “Image recoloring for color vision deficiency compensation,” Springer/IEEE-cited journal, 2021.
- [15] A. O. Eghari et al., “Testing a popular smartphone application for colour vision assessment in healthy volunteers,” *Clin. Ophthalmology*, 2021.
- [16] “Vector quantization-based color blindness test images,” *Proc. IEEE/ACM Int. Conf.*, 2017.
- [17] “Deep Learning Color Correction for Color Blindness (DeepCorrect),” Conf. paper, 2018–2020.
- [18] C. Bonanomi et al., “App-based assessment of SiChaRDa, an image enhancer for CVD,” *IEEE Imaging Symp.*, 2017.
- [19] A. Wicaksono et al., “Investigation of EEG signal response using event-related potentials with Ishihara stimuli,” *IEEE SPMB*, 2020.
- [20] H. Y. Lin et al., “Improving discrimination in color vision deficiency by image recoloring,” *Sensors*, 2019.
- [21] A. Padilla et al., “Realtime recoloring Ishihara plates using artificial neural networks,” *Proc. IEEE*, 2019.
- [22] R. S. B. Thapa et al., “Comparison of a smartphone application with Ishihara pseudoisochromatic plates for testing colour vision,” *Transl. Vis. Sci. Tech.*, 2019.
- [23] D. van Staden et al., “Comparing the validity of an online Ishihara colour vision test to the traditional handbook,” *African Vis. Eye Health*, 2018.
- [24] “A tunable digital Ishihara plate for preschool-aged children,” IEEE-adjacent conference, 2016–2018.
- [25] O. K. Ozgur et al., “Validity and acceptance of color vision testing on smartphones,” *J. Neuro-Ophthalmology*, 2018.

- [26] S. J. Dain and A. AlMerdef, “Colorimetric evaluation of iPhone apps for colour vision tests based on the Ishihara test,” *Clin. Exp. Optometry*, 2016.
- [27] M. E. Gordon et al., “Comparison of Ishihara booklet with color vision smartphone applications,” *Can. J. Ophthalmology*, 2016.
- [28] S. K. S. Naidu et al., “Re-coloring and wavelength adjustment algorithms for color vision deficiency,” in *Proc. IEEE Int. Conf. Inventive Systems and Control*, 2018.
- [29] R. Bove et al., “Evaluating more naturalistic outcome measures: A 1-year smartphone study in multiple sclerosis,” *Neurology*, 2015.
- [30] J. Ng et al., “Evaluation of the Waggoner computerized color vision test,” *Transl. Vis. Sci. Tech.*, 2015.
- [31] J. H. Stewart et al., “A modified protocol for color vision screening using Ishihara,” *Aviation, Space, Environ. Med.*, 2015.
- [32] S. J. Dain et al., “New iPad-based test for the detection of color vision deficiencies,” *Graefe’s Arch. Clin. Exp. Ophthalmology*, 2014.